

SBC 201 Chapter 13 Energy Efficiency — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise.
- **Deliverable tier:** Project-use matrices in Sections 1–5 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 13, source file Reference\SBC 201 2024\source_reference\Chapter_13 – ENERGY EFFICIENCY.txt.
- **Extraction audit:** Skill-finetune run. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold published tokens, building-language triggers, named exceptions, check-specific actions). Internal inventory: **0** independently checkable numeric records (**0** Verified, **0** Verify source). Chapter 13 has no tables, exceptions, footnotes, or numeric cells.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, energy-modelling report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from SBC 601, SBC 602, climate-zone tables, envelope/HVAC/lighting criteria, commentary examples, or the existing chapter summary. Where Chapter 13 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Chapter 13 does not assign **SBC 601** versus **SBC 602** to this occupancy. Both volumes are named; neither is selected here.
4. Automatic sprinkler protection, storey count, mixed-use podium and amenity program are unconfirmed. Chapter 13 itself does not branch on those facts; any such branching lives in the outbound energy codes and is not reconstructed here.
5. Climate zone, thermal envelope, air leakage, mechanical, service-water heating, electrical distribution and illumination criteria are not published in this chapter.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, use, occupant load, sprinkler branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 13 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 13 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 13 application	Required project action
SBC 601 versus SBC 602	Unconfirmed; Chapter 13 names both and does not assign either volume to Group R-2 high-rise	Controls which energy-conservation volume demonstrates compliance for the tower and any mixed-use portions	Energy/MEP consultant and AHJ to lock the applicable volume(s); do not infer the selection from this chapter
Climate zone	Not published in Chapter 13	Envelope, HVAC and related criteria in the outbound codes are climate-dependent	Establish the project climate zone from the locked SBC 601/602 documents, not from this matrix
Envelope, HVAC, service water, electrical and illumination criteria	Not published in Chapter 13	There are no R-values, SHGC, LPD, equipment-efficiency or similar numeric checks in this chapter	Produce the energy-compliance package from SBC 601 and/or SBC 602; keep those values off this matrix
Mixed-use / podium program	Unconfirmed	Shared or non-residential portions, if present, are still demonstrated under the outbound energy codes, not under a Chapter 13 occupancy table	Freeze occupancy/use by space; apply the locked 601/602 scoping to each portion
Energy consultant / AHJ pathway	Unconfirmed	Chapter 13 is a charging pointer only; it cannot evidence energy-code acceptance	Engage the qualified local energy/MEP consultant before design freeze

4. Energy-efficiency charging

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1301.1	Chapter energy-efficiency scope	This chapter governs the design and construction of buildings for energy efficiency	All buildings in Chapter 13 scope	None stated	Direct	Place Chapter 13 on the code sheet as the charging energy-efficiency chapter	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1301.1.1	SBC 601/602 compliance	Buildings shall be designed and constructed in accordance with the Saudi Energy Conservation Code (SBC 601 and SBC 602)	All buildings governed by this chapter	None stated	External verification	Demonstrate energy-efficiency compliance through SBC 601 and SBC 602; do not adopt envelope, HVAC, lighting or climate-zone values from this chapter	Verified

5. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger and branch are confirmed.
2. There is no **Verify source** row in this deliverable. Do not treat the absence of numeric cells as permission to fill values from memory, commentary or the chapter summary.
3. Do not import SBC 601 or SBC 602 envelope, HVAC, service-water, electrical-distribution, illumination or climate-zone figures into drawings from this matrix.
4. Do not treat Chapter 13 as assigning **SBC 601** or **SBC 602** to Group R-2 high-rise.
5. Record the locked energy-code volume(s), climate zone and compliance path in the project Golden Thread; this matrix is not evidence of SCD NOC, SBPS approval or stamped energy compliance.

6. Coverage summary

Internal inventory of the attached Chapter 13 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 0
- **Verified:** 0
- **Verify source:** 0

Counts by top-level section

Top-level section	Records
1301	0

Coverage cross-check against SBC 201 Chapter 13 Energy Efficiency (2024)_CS.md was topics-only: Chapter 13 is a charging pointer to **SBC 601** and **SBC 602**; it is not a thermal-envelope or lighting-power chapter. No CS.md value was copied into a matrix cell.

7. Unresolved-source register

No OCR, flattened-table, page-split or footnote hold point. Mandatory text is unambiguous. Numeric energy criteria are outbound and are not adopted from this chapter.

Affected table / clause	Why unverified	Control note
—	No independently checkable numeric cell, table or footnote in the attached extract	Do not reconstruct SBC 601/602 values here; lock those criteria from the outbound codes